Homework 28.10.26 – Exercises 29–30

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Exercise 29

Let $f = \frac{1}{n}\sum_{i=1}^{n} f_i$, and assume that each f_i is convex and L_i-smooth. Define a gradient estimator by setting $g(x) = \nabla f_i(x)$, where the index i is chosen uniformly at random. Establish a bound of the form

$$G(x,y) \le 2A\, D_f(x,y) + C,$$

and specify what A and C are. Use this to derive a convergence result for SGD using this gradient estimator.

Proof

By the definition of $G(x,y)$ and the fact that $i \sim \mathrm{Uniform}(\{1,2,\dots,n\})$,

$$G(x,y) = \mathbb{E}_i\big[\|g(x) - \nabla f(y)\|^2\big] = \frac{1}{n}\sum_{i=1}^{n} \|\nabla f_i(x) - \nabla f(y)\|^2 .$$

Using $\|a+b\|^2 \le 2\|a\|^2 + 2\|b\|^2$ (Exercise 24) we get

$$\begin{aligned} G(x,y) &= \frac{1}{n}\sum_{i=1}^{n} \big\|\big(\nabla f_i(x) - \nabla f_i(y)\big) + \big(\nabla f_i(y) - \nabla f(y)\big)\big\|^2 \\ &\le \frac{2}{n}\left(\sum_{i=1}^{n} \|\nabla f_i(x) - \nabla f_i(y)\|^2 + \sum_{i=1}^{n} \|\nabla f_i(y) - \nabla f(y)\|^2\right). \end{aligned}$$

Since each f_i is convex and L_i-smooth, we have $\|\nabla f_i(x) - \nabla f_i(y)\|^2 \le 2L_i\, D_{f_i}(x,y)$. Hence, with $L_{\max} := \max_i L_i$,

$$\begin{aligned} G(x,y) &\le \frac{4}{n}\sum_{i=1}^{n} L_i\, D_{f_i}(x,y) + \frac{2}{n}\sum_{i=1}^{n} \|\nabla f_i(y) - \nabla f(y)\|^2 \\ &\le 4L_{\max}\cdot\frac{1}{n}\sum_{i=1}^{n} D_{f_i}(x,y) + \frac{2}{n}\sum_{i=1}^{n} \|\nabla f_i(y) - \nabla f(y)\|^2 \\ &= 2\,(2L_{\max})\, D_f(x,y) + C(y), \end{aligned}$$

where we used $D_{f_i}(x,y) \ge 0$ (convexity of f_i), the linearity $D_f = \frac{1}{n}\sum_{i=1}^{n} D_{f_i}$, and

$$C(y) := \frac{2}{n}\sum_{i=1}^{n} \|\nabla f_i(y) - \nabla f(y)\|^2 = 2\,\mathrm{Var}\big[g(y)\big].$$

As a result, we get

$$A = 2L_{\max}, \qquad C = C(y) = \frac{2}{n}\sum_{i=1}^{n} \|\nabla f_i(y) - \nabla f(y)\|^2 .$$

Convergence of SGD

First of all, it is easy to notice that the estimator is unbiased:

$$\mathbb{E}_i\big[g(x)\big] = \frac{1}{n}\sum_{i=1}^{n} \nabla f_i(x) = \nabla f(x).$$

Abd if we take $x = x^k$ and $y = x^\star$ in the bound above, and use the fact that $\nabla f(x^\star) = 0$, then we get

$$G^k = \mathbb{E}\big[\left\|g^k - \nabla f(x^\star)\right\|^2 \mid x^k\big] \leq 2(2L_{\max})\, D_f(x^k, x^\star) + \frac{2}{n}\sum_{i=1}^{n} \left\|\nabla f_i(x^\star)\right\|^2,$$

i.e. the AC assumption holds with

$$A = 2L_{\max}, \qquad C = 2\sigma_\star^2, \qquad \sigma_\star^2 := \frac{1}{n}\sum_{i=1}^{n} \left\|\nabla f_i(x^\star)\right\|^2.$$

If in addition f is μ-convex, Theorem 33 gives that for every stepsize $0 < \gamma \leq \frac{1}{2L_{\max}}$ the iterates of SGD satisfy

$$\mathbb{E}\big[\left\|x^k - x^\star\right\|^2\big] \leq (1-\gamma\mu)^k \left\|x^0 - x^\star\right\|^2 + \frac{2\gamma\sigma_\star^2}{\mu}.$$

So SGD converges linearly up to a neighbourhood of radius $\frac{2\gamma\sigma_\star^2}{\mu}$, which shrinks as $\gamma \to 0$. In particular, if $\nabla f_i(x^\star) = 0$ for all i (interpolation), then $C = 0$ and SGD converges linearly to $x^\star$.

Exercise 30 (Code)

Code up the SGD method described in the previous example on an optimization problem of your choice. Perform a sequence of numerical experiments testing various parts of the established theory.

What we have to do

We have to implement SGD

$$x^{k+1} = x^k - \gamma \nabla f_{i_k}(x^k), \qquad i_k \sim \text{Uniform}(\{1, \ldots, n\}),$$

with the gradient estimator from Exercise 29, and check numerically what Theorem 33 together with Exercise 29 predicts:

1. for $0 < \gamma \leq \frac{1}{A} = \frac{1}{2L_{\max}}$ we have $\mathbb{E}\big[\left\|x^k - x^\star\right\|^2\big] \leq (1-\gamma\mu)^k \left\|x^0 - x^\star\right\|^2 + \frac{\gamma C}{\mu}$;
2. SGD converges linearly only up to a neighbourhood whose radius is proportional to γ.

The problem

We take $n = 12$ one-dimensional quadratics

$$f_i(x) = a_i x^2 + b_i x, \qquad f = \frac{1}{12}\sum_{i=1}^{12} f_i.$$

We do not add constant terms c_i (i.e. we take $c_i = 0$), because a constant changes neither the gradients nor the minimiser, so it plays no role in the minimisation.

The coefficients are just numbers that we generate; there is nothing mathematically important about them. We only want easy numbers and easy functions that are simple to analyse. We take random integers $a_i \in \{1, \dots, 20\}$ and $b_i \in \{-20, \dots, 20\}$ (seed 0):

$$a = (18, 13, 11, 6, 7, 1, 2, 1, 4, 17, 13, 19),$$
$$b = (0, 4, 19, 9, 5, 2, 2, 18, -9, 13, 7, -20).$$

For such functions every quantity of the theory can be computed by hand. Let $\bar{a} := \frac{1}{n}\sum_i a_i$ and $\bar{b} := \frac{1}{n}\sum_i b_i$, so that $f(x) = \bar{a}x^2 + \bar{b}x$.

Smoothness. Since $f_i'(x) = 2a_i x + b_i$, we have $|f_i'(x) - f_i'(y)| = 2a_i|x - y|$, so f_i is L_i-smooth with $L_i = 2a_i$, and

$$L_{\max} = 2\max_i a_i, \qquad A = 2L_{\max} = 4\max_i a_i.$$

Each f_i is convex because $f_i'' = 2a_i > 0$.

Strong convexity. $f''(x) = 2\bar{a}$, hence f is μ-convex with $\mu = 2\bar{a}$.

The minimiser. $f'(x) = 2\bar{a}x + \bar{b} = 0$ gives

$$x^\star = -\frac{\bar{b}}{2\bar{a}} = -\frac{\sum_i b_i}{2\sum_i a_i}.$$

The constant C. From Exercise 29, $C = 2\sigma_\star^2$ where $\sigma_\star^2 = \frac{1}{n}\sum_i |f_i'(x^\star)|^2$, and $f_i'(x^\star) = 2a_i x^\star + b_i$, so we get

$$C = \frac{2}{n}\sum_{i=1}^{n}\big(2a_i x^\star + b_i\big)^2.$$

Our numbers. $\sum_i a_i = 112$, $\sum_i b_i = 50$ and $\max_i a_i = 19$, so

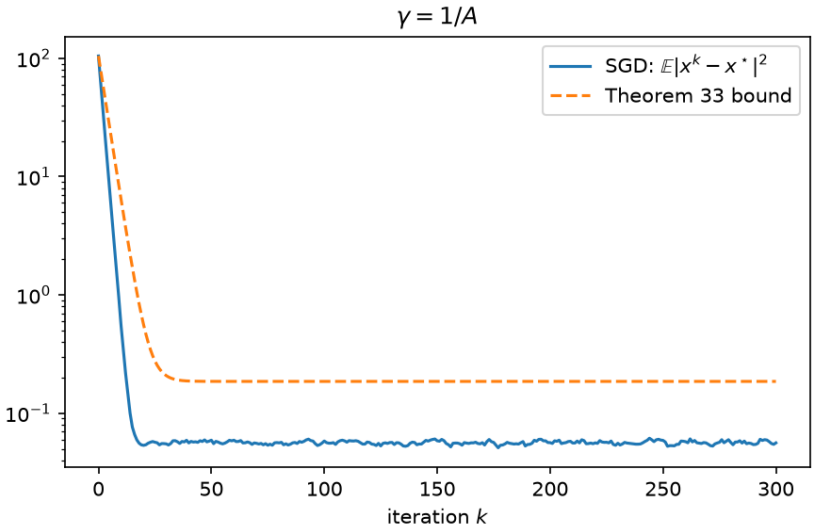
$$x^\star = -\tfrac{50}{224} = -\tfrac{25}{112} \approx -0.2232, \qquad \mu = \tfrac{112}{6} = \tfrac{56}{3} \approx 18.67,$$
$$L_{\max} = 38, \qquad A = 76, \qquad C \approx 263.31.$$

We start from $x^0 = 10$ and estimate $\mathbb{E}\big[\|x^k - x^\star\|^2\big]$ by averaging over 2000 independent runs of SGD.

The code is the file `Homework/Exercises/hw_2026_10_28/ex30_sgd.ipynb`.

Experiment 1: the bound of Theorem 33

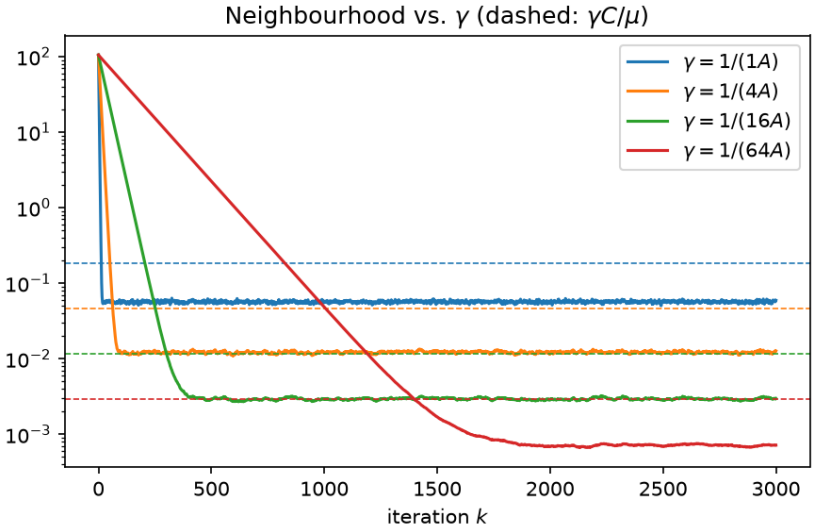
We take the largest stepsize allowed by the theory, $\gamma = \frac{1}{A} = \frac{1}{76}$.



The error decreases linearly and then stays at about $5.6 \cdot 10^{-2}$. The bound is never violated, and its limit is $\frac{\gamma C}{\mu} \approx 0.186$.

Experiment 2: the neighbourhood is proportional to γ

We take $\gamma = \frac{1}{mA}$ for $m = 1, 4, 16, 64$ (dashed lines: $\frac{\gamma C}{\mu}$).



γ	plateau (SGD)	exact limit	$\gamma C/\mu$
$1/A$	$5.67 \cdot 10^{-2}$	$5.67 \cdot 10^{-2}$	$1.86 \cdot 10^{-1}$
$1/(4A)$	$1.22 \cdot 10^{-2}$	$1.22 \cdot 10^{-2}$	$4.64 \cdot 10^{-2}$
$1/(16A)$	$2.94 \cdot 10^{-3}$	$2.93 \cdot 10^{-3}$	$1.16 \cdot 10^{-2}$
$1/(64A)$	$7.16 \cdot 10^{-4}$	$7.27 \cdot 10^{-4}$	$2.90 \cdot 10^{-3}$

Dividing γ by 4 divides the plateau by about 4, as predicted. The price is a slower linear rate $1 - \gamma\mu$. The plateau agrees with the exact formula, and the ratio plateau/$(\gamma C/\mu)$ tends to $\frac{1}{4}$.

Conclusion

The predictions of Exercise 29 and Theorem 33 are confirmed: SGD converges linearly up to a neighbourhood of radius at most $\frac{\gamma C}{\mu}$, and this neighbourhood shrinks proportionally to γ. The bound is about 4 times larger than the true neighbourhood.